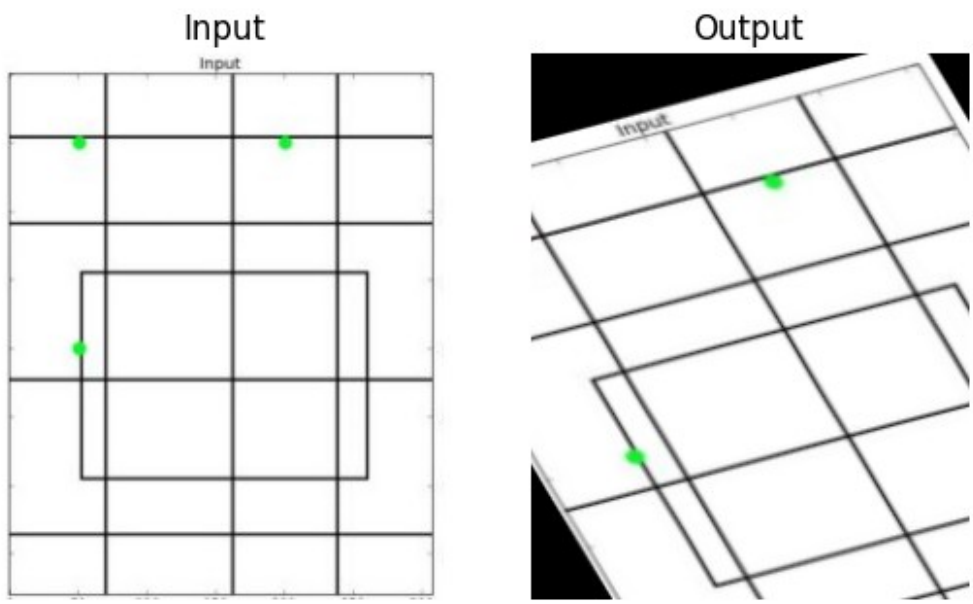
```
import numpy as np
import cv2 as cv
import matplotlib.pyplot as plt

img = cv.imread('C:/Users/sreev/Downloads/PROGRAM2/messi5.jpg')
assert img is not None, "file could not be read, check with
os.path.exists()"

rows, cols = img.shape[:2]

M = np.float32([[1, 0, 100],
                [0, 1, 50]])

dst = cv.warpAffine(img, M, (cols, rows))

# Convert BGR → RGB for correct colors
dst_rgb = cv.cvtColor(dst, cv.COLOR_BGR2RGB)

plt.imshow(dst_rgb)
plt.axis('off')
plt.title("Affine Transformed Image")
plt.show()
```

```
import cv2 as cv
import matplotlib.pyplot as plt
```

```
img =
cv.imread('C:/Users/sreev/Downloads/PROGRAM2/messi5.jpg',cv.IMREAD_GRA
YSCALE)
assert img is not None, "file could not be read"
rows, cols = img.shape
M = cv.getRotationMatrix2D(((cols-1)/2.0, (rows-1)/2.0), 90, 1)
dst = cv.warpAffine(img, M, (cols, rows))
plt.imshow(dst, cmap='gray')
plt.axis('off')
plt.show()
```

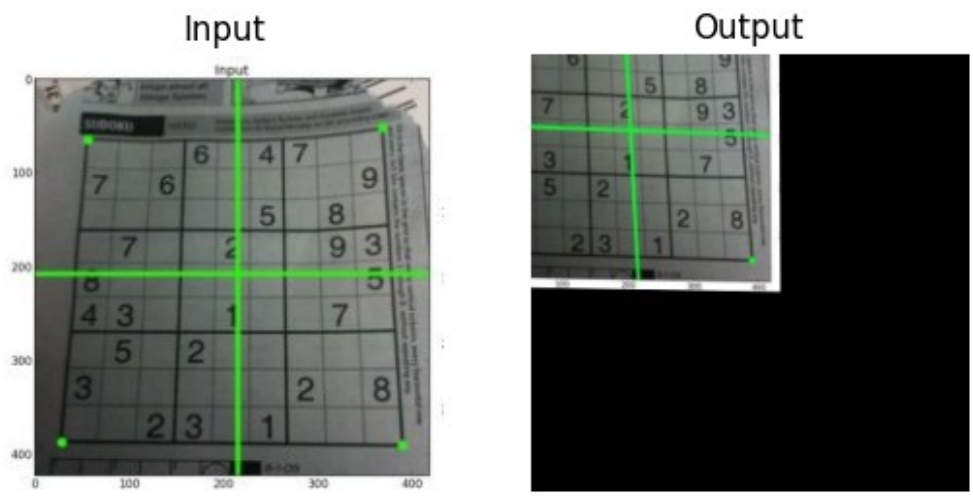
```
import cv2 as cv
import numpy as np
import matplotlib.pyplot as plt

img = cv.imread("C:/Users/sreev/Downloads/PROGRAM2/drawing.pnG")
assert img is not None, "file could not be read"

rows, cols, ch = img.shape

pts1 = np.float32([[50, 50], [200, 50], [50, 200]])
pts2 = np.float32([[10, 100], [200, 50], [100, 250]])

M = cv.getAffineTransform(pts1, pts2)
dst = cv.warpAffine(img, M, (cols, rows))
```

```
plt.subplot(121)
plt.imshow(img)
plt.title('Input')
plt.axis('off')

plt.subplot(122)
plt.imshow(dst)
plt.title('Output')
plt.axis('off')

plt.show()
```




```
import cv2 as cv
import numpy as np
import matplotlib.pyplot as plt

img = cv.imread("C:/Users/sreev/Downloads/PROGRAM2/sudoku.png")
assert img is not None, "file could not be read"

rows, cols, ch = img.shape

pts1 = np.float32([[56, 65], [368, 52], [28, 387], [389, 390]])
pts2 = np.float32([[0, 0], [300, 0], [0, 300], [300, 300]])

M = cv.getPerspectiveTransform(pts1, pts2)
dst = cv.warpPerspective(img, M, (300, 300))


plt.subplot(121)
```

```
plt.imshow(img_rgb)
plt.title('Input')
plt.axis('off')

plt.subplot(122)
plt.imshow(dst)
plt.title('Output')
plt.axis('off')

plt.show()
```




```
import cv2
import numpy as np
import matplotlib.pyplot as plt
img = cv2.imread("C:/Users/sreev/Downloads/PROGRAM2/flower.jpg")
assert img is not None, "Image not found! Check the path."

img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
plt.figure("Image Transformations", figsize=(12, 10))
plt.subplot(3, 4, 1)
plt.imshow(img_rgb)
plt.title("Original Image")
plt.axis("off")
```

```
(np.float64(-0.5), np.float64(165.5), np.float64(147.5), np.float64(-
0.5))
```

Original Image

```
# 2. Cropping

cropped = img[50:300, 100:400]
cropped_rgb = cv2.cvtColor(cropped, cv2.COLOR_BGR2RGB)

plt.subplot(3, 4, 2)
plt.imshow(cropped_rgb)
plt.title("Cropped Image")
plt.axis("off")
```

```
(np.float64(-0.5), np.float64(65.5), np.float64(97.5), np.float64(-0.5))
```

Cropped Image

```
# 3. Resizing

resized = cv2.resize(img, (300, 300))
resized_rgb = cv2.cvtColor(resized, cv2.COLOR_BGR2RGB)
plt.subplot(3, 4, 3)
plt.imshow(resized_rgb)
plt.title("Resized Image")
plt.axis("off")
```

```
(np.float64(-0.5), np.float64(299.5), np.float64(299.5), np.float64(-0.5))
```

Resized Image

```
# 4. Translation

rows, cols = img.shape[:2]
M_trans = np.float32([[1, 0, 50],[0, 1, 30]])
translated = cv2.warpAffine(img, M_trans, (cols, rows))
translated_rgb = cv2.cvtColor(translated, cv2.COLOR_BGR2RGB)
plt.subplot(3, 4, 4)
plt.imshow(translated_rgb)
plt.title("Translated Image")
plt.axis("off")
```

```
(np.float64(-0.5), np.float64(165.5), np.float64(147.5), np.float64(-0.5))
```

Translated Image

```
# 5. Rotation

center = (cols // 2, rows // 2)
M_rot = cv2.getRotationMatrix2D(center, 45, 1.0)
rotated = cv2.warpAffine(img, M_rot, (cols, rows))
rotated_rgb = cv2.cvtColor(rotated, cv2.COLOR_BGR2RGB)
plt.subplot(3, 4, 5)
plt.imshow(rotated_rgb)
plt.title("Rotated (45°)")
plt.axis("off")
```

```
(np.float64(-0.5), np.float64(165.5), np.float64(147.5), np.float64(-0.5))
```

Rotated (45°)

```
# 6. Scaling

scaled = cv2.resize(img, None, fx=1.5, fy=1.5)
scaled_rgb = cv2.cvtColor(scaled, cv2.COLOR_BGR2RGB)
plt.subplot(3, 4, 6)
plt.imshow(scaled_rgb)
plt.title("Scaled Image")
plt.axis("off")
```

```
(np.float64(-0.5), np.float64(248.5), np.float64(221.5), np.float64(-0.5))
```

Scaled Image

```
# 7. Shearing

M_shear = np.float32([[1, 0.5, 0],[0, 1, 0]])
sheared = cv2.warpAffine(img, M_shear, (int(cols * 1.5), rows))
sheared_rgb = cv2.cvtColor(sheared, cv2.COLOR_BGR2RGB)
plt.subplot(3, 4, 7)
plt.imshow(sheared_rgb)
plt.title("Sheared Image")
plt.axis("off")
```

```
(np.float64(-0.5), np.float64(248.5), np.float64(147.5), np.float64(-0.5))
```

Sheared Image

```
# 8. Reflection (Flip)

flipped = cv2.flip(img, 1)
flipped_rgb = cv2.cvtColor(flipped, cv2.COLOR_BGR2RGB)
plt.subplot(3, 4, 8)
plt.imshow(flipped_rgb)
plt.title("Reflected (Flip)")
plt.axis("off")
```

```
(np.float64(-0.5), np.float64(165.5), np.float64(147.5), np.float64(-0.5))
```

Reflected (Flip)

```
# 9. Projective (Perspective) Transformation

pts1 = np.float32([[50, 50],[cols - 50, 50],[50, rows - 50],[cols - 50, rows - 50]])
pts2 = np.float32([[10, 100],[cols - 100, 50],
                   [100, rows - 10],
                   [cols - 50, rows - 50]])

M_persp = cv2.getPerspectiveTransform(pts1, pts2)
perspective = cv2.warpPerspective(img, M_persp, (cols, rows))
perspective_rgb = cv2.cvtColor(perspective, cv2.COLOR_BGR2RGB)

plt.subplot(3, 4, 9)
plt.imshow(perspective_rgb)
plt.title("Perspective Transform")
plt.axis("off")

plt.tight_layout()
plt.show()
```